

Freight Rate Prediction

ML Pipeline Report

1. Data Exploration and Key Findings

The development dataset contains 48,000 loads and 14 columns: identifiers, pickup/delivery information, coordinates, distance, equipment, weight, date, market/quote signals, and the target posted_rate.

Finding	Observation / treatment
Unique load IDs	48,000 unique IDs; no duplicate load_id values.
Missing values	300 missing weights and 374 missing market_index values. Neither missing field is required by the final feature set; CatBoost can also handle missing numeric values.
Negative weights	292 rows (0.61%) had negative weight values. They were treated as sign errors and converted with abs().
Equipment	Three categories: Dry Van, Reefer, Flatbed. equipment is modeled as a categorical feature.
Dates	Data is naturally ordered in time; dates run from January through October 2025.
Coordinates	Pickup and delivery city coordinate consistency was checked; inconsistent city-coordinate mappings would raise an error.
Distance	All observed distances are positive; distance is retained and expanded with nonlinear transforms.

2. Preprocessing and Data Quality

The preprocessing is deliberately conservative. Dates are parsed as datetime values and the full development data is sorted chronologically. Duplicate load IDs are checked. Pickup and delivery coordinate consistency is checked at the city level. The only explicit value correction is the sign of weight: negative values are converted to positive values because weight cannot be negative and the notebook's distribution check supports interpreting the negatives as sign errors.

The final model does not use market_index, quote_signal, or latitude/longitude fields. The final feature set is kept compact and consists of distance, distance transforms, weight, weight-per-mile, equipment, and date-derived features.

3. Feature Engineering

The current model uses the following features:

Feature	Purpose
distance	Direct measure of lane length.
distance_sq	Allows the model to capture nonlinear distance effects.
sqrt_distance	Provides a softer nonlinear distance transformation.
weight	Freight weight after sign correction.
weight_per_mile	Captures load intensity relative to lane length.
equipment	Categorical equipment type; handled natively by CatBoost.
dayofweek	Captures weekly seasonality.
is_weekend	Explicit weekend indicator.
day	Captures within-month/date-position effects.

The raw market_index, quote_signal, pickup/delivery city names and latitude/longitude fields are currently checked or available in the data but are not passed to the final model.

4. Train / Validation Strategy

A chronological split is appropriate because the final prediction set contains later dates than the labeled development data. The current implementation uses data before 2025-10-01 for model development and October 2025 as a final temporal hold-out. This avoids training on future observations when evaluating October.

Inside the pre-October development period, TimeSeriesSplit with 6 folds is used for model/iteration selection. Each validation fold occurs after its corresponding training window. The mean best iteration from the six folds is approximately 194, and the final model is trained with 193 iterations.

5. Model Choice and Training

The model is CatBoostRegressor with RMSE loss. The target is transformed with $\log_{1p}$ before training and converted back with $\expm1$ for evaluation and final predictions. This is useful for a positive, right-skewed rate target and reduces the influence of very large rates during training.

Parameter	Final value
Model	CatBoostRegressor
Target	$\log_{1p}(\text{posted_rate})$
Iterations	193
Learning rate	0.05
Depth	6
L2 regularization	5
Loss	RMSE
Random seed	42
Categorical feature	equipment

6. Validation Results

The project notebook records the following approximate mean cross-validation metrics and October hold-out metrics. The hold-out is the more important generalization check because it represents a future time period not used during model development.

Metric	6-fold CV (approx.)	October hold-out
MAE	146.27	120.18
RMSE	598.44	650.77
R^2	0.84	0.82
MAPE	6.30%	5.78%

The hold-out MAE and MAPE are strong and R^2 is around 0.82. RMSE is substantially higher than MAE, which indicates that a relatively small number of large errors/outliers remain. This is expected for a rate target with a long right tail.

7. Feature Importance

The fitted model's feature importance is dominated by distance-related variables. This is consistent with freight pricing: lane length is a primary driver of the expected rate. Equipment is the next most important categorical factor, while weight and the calendar variables have smaller but non-zero contributions.

Feature	CatBoost importance
distance_sq	33.44%
sqrt_distance	31.04%
distance	30.09%
equipment	3.32%
weight	0.92%
weight_per_mile	0.76%

day	0.21%
dayofweek	0.18%
is weekend	0.05%

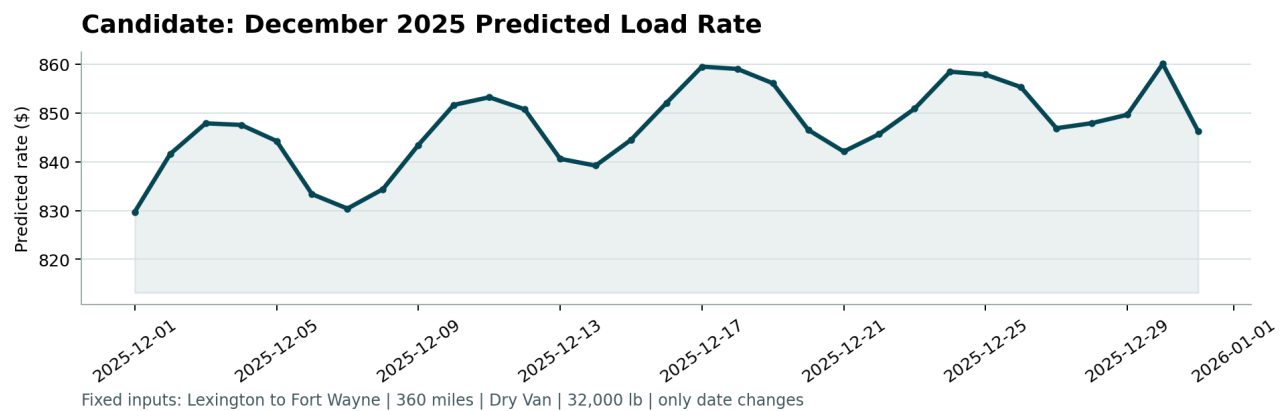
8. Final Prediction Generation

After the final model is trained on all pre-October development rows, the same preprocessing and feature engineering are applied to data/validation.csv. Predictions are generated for all 12,000 validation loads, rounded to two decimals, and written into validation-predictions-template.csv to produce validation_predictions.csv.

The completed validation file contains exactly two columns, load_id and predicted_rate, with 12,000 unique IDs matching the required TE-000001 through TE-012000 range.

9. Fixed December Prediction and Scoring

The provided score.py was executed successfully and generated the required candidate_december.png. The December scenario keeps pickup, delivery, distance, equipment and weight fixed and changes only the date.



December predictions range from \$829.73 to \$860.08, with an average of \$847.32.